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56) TASK #5: Space Base Automation:

Objective: Build & program a working prototype that simulates the automated of 3 systems in a logical sequence to protect what your team deems the most important assets as power decreases.

1) Atmosphere & Life Support
- ~~[illegible]~~ O_2 Generator
- Air Circulation & scrubbing
- Dehumidifier ventillation

2) Water Management
- Recycling Pumps
- Water treatment & filtration
- Waste Unit Processing

3) Food Production:
- Greenhouse Grow lighting
- Irrigation & nutrient pumps
- Cold storage refrigeration

4) Structure & Monitoring
- Pressure sensor & alarms
- Airlock Control Systems
- Interior lighting

5) Health & wellbeing
- Medical Bay Equipment
- Exercise turbine Generators
- Earth Simulator AV systems

6) Communications
- Primary Antenna Transmitter
- Backup Antenna Transmitter
- PC Setup

7) Chemical treatment
- Sabatier Reactor
- Electrolysis of water
- Thermal control system (ACTS)

Chosen

Shutdown Order:

1) ACTS Thermal control system:
- Heat Distribution is the most noncritical as the body heat of the crew as well as the aerogel insulation should keep the temperature relatively same for short durations.

2) Sabatier Reactor:
- This produces methane gas for power & extra water, but both of these are produced in other ways meaning its shut off can be accounted for.

3) Electrolysis of Water
- This is the primary source of O_2 making it an absolute maximum priority and it should be shut off last.

Task 1 Reflection
- In the ~~[illegible]~~ Tinker Cad program, the maximum output of the solar panel was 1000 but in my room it was only around ~280, so I had to address the thresholds, but after that the process was smooth.

Sunesh

Team Name: TMI
Major System: Chemical Treatment
Team Members: James Harper, Rishi Reddy, Belal A, Basel A
Participation: Asynchronous w/ Friend

As our system's power decreases, we sequentially shut off systems in increasing priority. The least critical systems, like entertainment or lights will shut off first. In terms of code, my code changed from just reading the analog value to also sending it.

We encountered a couple issues. 1st, in my conditional that checks if Level = 3, this is syntactically incorrect, and would set the variable level = 3, rather than checking if level is = 3. Thus, regardless of the light, the system would always stay on.

During integration, we encountered another bug where clients on the receiving end would end up receiving ~~the~~ almost random numbers. This was because I was printing the level variable for debug purposes, which would append an ascii value to the transmitted hexadecimal value.

Overall, I think it went pretty well; we coordinated our systems. I think if we spent more time making sure our code would be compatible before we combined them, it would have been more efficient.

#Reflection:
I think a more customizable system would analyze the voltage & wattage provided by the existing power & compare it with the requirements by different components in a master list to decide what it should run out of what it can run.

James H